COSC2671 — Social Media and Network Analytics

Assignment 2: The Sudden Fuel Price Hike

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1. Introduction

Prices for petrol and diesel are rarely stagnant for long in Australia; between geopolitical shocks, fluctuations in currency, changes in refining margins and the domestic fuel cycles there is rarely a long time that either the price at the browser or a related, highly publicized consumer cost indicator doesn't shift. In May 2026 another bout of fuel price increases - largely fueled by escalating conflict in the Middle East and subsequently having a knock on effect upon global oil benchmarks - saw the kind of public furore that has become a regular mainstay of social media feeds. In this report we aim to analyse this furore on YouTube, the longest-form and highest commenting-density social platform available on the planet.

The research question that guides this study is: "What does discourse around the May 2026 fuel price hike on YouTube represent regarding public sentiment, what major topics are discussed, and where does the discussion occur?". It is purposefully three-pronged, each facet requiring a distinct approach to analysis. Sentiment analysis requires quantification of the emotional undercurrents of discourse, topic modelling will identify the underlying subjects that are actively debated, and network analysis will provide clarity over which particular channels and groups of commenters have become the nodes and foci of such discourse.

YouTube was chosen as a data source due to three main factors. Firstly, comments attached to news and explainers tend to be longer and more discursive than other short-form social media comments or video reactions, thus providing a significant advantage for sentiment scoring and topic modelling. Secondly, YouTube comments maintain a strong connection to a concrete channel and video, which greatly simplifies the construction of a directed network showing user interaction, a difficulty encountered with platforms where such connections are weaker or non-existent. Thirdly, YouTube is much less subject to automated comments than sites such as X (previously known as Twitter);

The report follows a general structure report-in essence of a single narrative flowing from the collection and cleaning of data to exploratory statistics, sentiment analysis, topic modelling, and network analysis. This report, however, constitutes the first half of the process, running from the initial collection phase up to preliminary qualitative findings that will inform subsequent more advanced analyses.

2. Background and Motivation

Fuel price hikes are not so much an economic indicator as a pain point at the household level, and that pain looks a lot the same whether you are in Sydney, Mumbai, or Houston. Petrol typically makes up between two and five percent of weekly household expenditures in developed and developing economies, but in the outer suburbs, rural areas or transit-reliant pockets of a nation where driving is not an option, this figure will easily double. Those with the least capacity to shoulder the price increase are therefore often hit hardest by it, and this asymmetry is loud and clear in social media discussion during and immediately following each hike. An indicator such as cash rate or trade balance takes effort to understand, but the number on a service station sign doesn’t. You see it, remember it, and talk about it. It's also one of the few items of cost-of-living information that appears to provoke the same kind of reaction regardless of country, which makes it a rich topic of social media analysis.

The 2026 price rise, however, is a shock superimposed onto a string of others, including the post-pandemic price surge of 2022, the September 2023 OPEC+ supply cuts, and numerous minor increases stemming from Red Sea disruptions. The defining feature of the 2026 spike among the data studied here is the intensity and abruptness of the increase. For about three years prior, average monthly commentary concerning fuel price on YouTube barely nudged beyond twenty to a hundred remarks; in the March, April and May of 2026, the figure was thousands per month. It is this sort of step change event detection methods are made to spot, signaling a single event rather than a slow background mumble.

There is one caveat on the dataset that must be clarified before proceeding with the analysis, for it underlies everything which comes after. YouTube shows return search results irrespective of origin. This set comprises content retrieved from mainstream Australian news sources (9 News Australia, ABC News Australia, 7NEWS Australia, Sunrise) alongside major US providers (MS NOW, CNN, NBC News) and a significant portion of Indian fuel price content (India Today, ABP NEWS, News18 India, CNBC Awaaz, Aaj TV, GTV Network). The mix of locations here is considered not an anomaly to be corrected for, but rather a characteristic of the topic itself, for fuel price hikes are, in their nature, transnational phenomena whose ensuing discourse does not discriminate between borders. All three analytical approaches developed below (sentiment, topic and network analysis) will thus be carried out on the entire aggregated dataset.

Using sentiment analysis (VADER) in tandem with LDA topic modeling and centrality-based network analysis essentially provides three perspectives on a single phenomenon: sentiment reveals the emotional tenor of discussion, topics illuminate the content, and network shows the dynamics of discourse. Fuel price discourse is analyzed through this same three-pronged approach.

3. Data Collection

All data was collected via the YouTube Data API v3 using a custom Python pipeline organised around three helper modules (youtube_collector.py, network_builder.py, and nlp_utils.py) by a dedicated Jupyter notebook The pipeline performs four sequential operations: video search by keyword, video statistics retrieval, comment thread retrieval, and channel metadata retrieval. Each operation is wrapped in error-handling logic that allows the pipeline to continue if a particular video has comments disabled or a particular channel returns malformed metadata — both of which occur regularly enough that ignoring them would have caused significant data loss.

3.1 Search Strategy and Query Set

Eight search queries were used to maximise coverage of the fuel price hike discourse without restricting the dataset to a single phrasing. The queries combine event-focused, reaction-focused, and geography-focused framings, and were drafted to reflect the kinds of phrases that both producers and consumers of fuel-price content would actually type.

Search Query	Framing / Intent
fuel price hike 2026	Event-anchored, time-bounded
petrol price increase 2026	Synonym-anchored, time-bounded
gas price spike reaction	Reaction-focused (US terminology)
fuel cost crisis people reaction	Reaction-focused, sentiment-implying
petrol price hike australia 2026	Geography-anchored (Australia)
oil price surge economy impact	Macro-economic framing
fuel prices too high opinion	Opinion / vox-pop framing
why is petrol so expensive 2026	Question-style, search-natural

For each query, up to thirty videos were retrieved (the YouTube API permits a maximum of fifty per call, but thirty were selected to leave headroom under the daily quota), filtered to English language results ordered by relevance, and bounded by a published_after date of 1 January 2023. This three-year-plus window was deliberate: a long lookback period was needed to establish a credible baseline of normal commentary volume against which the 2026 spike could be measured.

3.2 API Quota Constraints

The Daily Quota is set to 10,000 for the YouTube Data API v3. Different endpoints have vastly different values: the endpoint search.list costs 100 units, whereas videos.list and commentThreads.list consume just one unit each (videos list returns up to 50 videos per request, whereas commentThreads.list returns 100 comments). In the case of this particular analysis, the major costs were the search part (8 calls × 100 = 800 units), followed by per-video comments extraction (240 calls × 1 = 240 units). The overall quota cost was approximately 1,300 units for one iteration of collecting data. Moreover, a small delay (sleep of 0.3 second) between calls to the comment threads endpoint was implemented to not trigger any rate-limiting warnings – the standard practice when working with YouTube API. Compared to limitations on Twitter API, which were rather strict, especially regarding limit on tweet retrieval, the quota was more than generous.

3.3 Final Dataset Composition

Following the process of deduplication for video_id and comment_id, the dataset finally collected consists of 212 distinct videos in 136 distinct channels, comprising 11,716 comments published within a date range of 2 January 2023 to 18 May 2026. The aggregate engagement of the set of videos is quite high: total number of views on all 212 videos is at a staggering 48.25 million, an average of 227,616 views per video, and there is one peak video, from the Auto Reviews channel regarding the newly launched Toyota Land Cruiser (which was identified using the general "fuel cost" query rather than specifically looking for fuel price videos), viewed 8.15 million times. However, the median number of views is only 23,722, indicating that long tail phenomenon which we explore further in section 5.

Dataset Element	Final Count
Search queries	8
Unique videos collected	212
Unique channels (content producers)	136
Raw comments collected	11,716
Total video views across the set	48,254,509
Date range of video publication	Jan 2023 – May 2026
Approximate API quota consumed	~1,300 units / 10,000 daily

3.4 Channel Interaction Network Construction

Besides the tables containing information about video and comment features, there are two different types of networks generated based on the collected data in the notebook titled 03_network_analysis. One of them is an undirected and weighted user-co-comment network, where two users are considered neighbors if they comment on at least two common videos, and the weight between any pair of users is the number of common videos. The other one is the directed and weighted channel influence network, where the direction goes from the commenter to the target channel hosting the video that the commenter interacted with. The direction and weight are defined as the number of interactions, and the channel_indegree.csv in the dataset (10,624 records) reflects the inbound side of this graph.

4. Pre-processing and Data Cleaning

The preprocessing phase in 02_preprocessing.ipynb consists of generating three derived artifacts out of the raw data – videos processed in the videos table (videos_processed.csv), the comments table that has been fully processed but keeps the non-English comments for the sake of network analysis, and the filtered comments table . The decision to generate three artifacts instead of reducing everything to one artifact was intentional since while the network analysis requires all comments it can find, sentiment and topic analysis require a thorough filtration of all the irrelevant comments.

4.1 Videos Table Cleaning and Feature Engineering

The processing of the videos table was quite straightforward. The numeric statistics columns were parsed as integers, with any error set to 0, the published_at column parsed to UTC timezone-aware timestamp, and any entry without video_id value or not having a parseable publication date was removed. Duplicate entries based on the video_id key were dropped, however, since this operation was performed in the collection process, dropping duplicates served more as verification rather than an essential cleaning step. Also, three time features, which will be used in the temporal analysis in Section 5, were created – year, year_month and day_of_week.

To enable ranking and grouping of videos by overall user engagement, a feature was engineered which incorporates views, likes and comments of the video, weighted by the perceived intentional effort needed for each activity (clicking to view video is least intentional; liking involves deliberate click; commenting takes up some time to type message). The result was a min-max scaled to the [0,1] range of view_count*1 + like_count*10 + comment_count*20 engagement score, which was further grouped in quartiles (Low, Medium, High and Viral), forming a classic long tail – out of 212 videos, the Low group contains 206 videos, while the other three one High, and two Viral categories exist.

4.2 Comments Cleaning Pipeline

A much more substantive approach was required for cleaning up the comments dataset. These are the steps taken by the data pipeline to process each row of data: discard rows where comment_text has no content or only consists of whitespace; discard rows where comment_text contains less than ten characters; keep unique entries based on the comment_id column; convert the published_at column to a UTC timestamp, dropping unconvertible rows; convert the like_count and reply_count columns to integers; and create the year_month, day_of_week, and hour-of-day columns as features. A filter was then implemented to identify English-language posts. This consisted of classifying comments as English if more than seventy-five per cent of the characters in comment_text were ASCII-encoded.

In order to prepare the slice for use in topic modelling, there were an additional three filters applied. Firstly, any exact duplicates in the text were found and removed, which catches the most obvious form of text that would appear via copy-paste comment spamming. Secondly, all commenters that posted more than twenty comments in total across the entire dataset were identified as prolific commenters and subsequently removed – the effect of this is to conservatively filter out the very few users that could be thought of as acting like promoter bots. Finally, comments below three tokens after stopwords had been removed were excluded as being insufficiently long to enable topic assignment. The actual text went through a normalization process involving a cleaning function that involved lowercase processing, URL removal, removal of '@' mentions and '#' tags, removal of non-alphabetic characters apart from light punctuation and collapsing of whitespace. The tokens were processed using WordNet Lemmatizer and filtered through a combination stopword list that consisted of NLTK default English stopwords, along with a domain-specific stopword list.

This domain stopword list is particularly important to highlight, because its inclusion improves the quality of topic modelling by a significant amount. Words like 'fuel', gas, petrol, price, prices, oil, cost, hike, pump, litre, gallon, per were removed since these words occurred in all the documents, thus overwhelming every topic. As in the case of the RUOK day report where custom domain stop words were created along with the existing NLTK ones to improve topic modelling accuracy, similar actions have been performed here as well.

4.3 Attrition Across the Pipeline

Tracking how many records survive each stage of the pipeline is important both for transparency and for interpreting downstream results. Table 3 below summarises the attrition.

Stage	Comments In	Removed	Comments Out
Raw collected from API	—	—	11,716
Null / short / duplicate removal	11,716	318	11,398
English-only filter (75% ASCII heuristic)	11,398	410	10,988
Spam filter (duplicate text)	10,988	~700	~10,300
High-volume poster filter (>20 comments)	~10,300	~600	~9,700
Token count < 3 after stopword removal	~9,700	~200	9,509 (NLP-ready)

There are two artifacts produced at this step. The first is the comments_processed.csv which keeps all the 11,398 de-duplicated comments inclusive of non-English text, which is the data that will be used for the channel level network analysis where all interactions matter and irrespective of whether they are in English or not. The second is the comments_nlp_ready.csv that holds the 9,509 comments which have been filtered and tokenized, as well as converted to English only for use as inputs in the VADER and LDA models.

5. Initial and Qualitative Insights

Before moving on to sentiment scoring, topic modeling, and social network analysis, a shape of the data understanding should first be built through descriptive exploration. Here follows an example of such a descriptive exploration performed using five perspectives: engagement analysis, leaderboard analysis of the channel and videos, analysis of temporal behavior both at a month-by-month scale and on an intra-day basis, and a quick vocabulary exploration.

5.1 The Long-Tail of Engagement

The Distribution of engagement among the 212 analyzed videos has a clear long-tailed profile. The average number of video views is 227,616 whereas the median is much lower – 23,722. That's the simple numerical proof that we deal with a long-tail phenomenon; there is the same pattern concerning video likes – average likes amount to 7,623 while the median is only 178; as for comments, their average number is 428 while median equals 26. In terms of the defined engagement levels, that means that out of 212 considered videos, 206 belong to the Low level category, three videos – to the Medium level, one video belongs to the High level while two videos belong to the Viral level.

From the practical perspective, it matters because there are two implications to the following analysis of videos. First, summarization of sentiment or topics by averaging videos would result in biased results due to dominating low-engagement level. Second, two Viral videos and their hosting channels would most likely become the most central nodes, and therefore the interpretation should consider that their high centrality level is mostly associated with video engagement rather than influence.

5.2 Channels Driving the Conversation

Examining which channels produced the most fuel-price content in the collected window gives a useful first read on who is shaping the supply side of the discourse. The top ten channels by video count are split almost evenly between Australian news outlets and international news outlets, with a strong contingent of Indian news channels reflecting the parallel fuel-price discourse in that market.

#	Channel	Region	Videos
1	9 News Australia	Australia	11
2	MS NOW	United States	10
3	ABC News (Australia)	Australia	8
4	CNN	United States	8
5	India Today	India	7
6	7NEWS Australia	Australia	5
7	NBC News	United States	4
8	ABP NEWS	India	3
9	Sunrise	Australia	3

#	Channel	Region	Videos
10	News18 India	India	3

The four leading producers among the top ten channels are Australian mainstream news networks (9 News Australia, ABC News Australia, 7NEWS Australia, Sunrise). These channels altogether made up 27 out of the total number of 212 videos, or 13 per cent. The other three are Indian news networks (India Today, ABP NEWS, News18 India). There are three American news networks (MS NOW, CNN, NBC News). The most productive channel, the one that has uploaded the highest number of videos in the dataset, is 9 News Australia, with 11 videos in total, perfectly fitting into the category of news cycle media content by posting news almost daily on an ongoing story about the cost of living. Having three Indian news networks among the top ten clearly shows that the dataset, even though centered on Australia, is more of a world discussion issue.

5.3 Top Videos by Engagement

The most-viewed videos in the collected set are a mix of fuel-price news content and adjacent automotive content surfaced by the broader queries. The top five by view count are listed below.

Title (truncated)	Channel	Views	Comments
The new Toyota Land Cruiser is too cool!	Forrest's Auto Reviews	8,147,520	3,648
Why don't we have more nuclear power plants?	Hank Green	6,651,820	17,752
Should You Buy An Electric Car (Shorts)	Nikhil Kamath	4,841,865	1,614
Petrol Price To Be Rs.15/L? – Nitin Gadkari	CNBC Awaaz	3,448,164	8,358
PETROL vs DIESEL – Warikoo Shorts	warikoo	2,957,649	860

Two points can be highlighted in relation to the above observations. Firstly, the top viewed video on fuel prices proper – that is CNBC Awaaz's Hindi language Shorts video on the proposal by the Indian Transport Minister Nitin Gadkari for an ethanol blended fuel priced at fifteen rupees per litre – occupies the fourth position on the table overall but ranks as the video on fuel prices with the most number of comments (8,358). The explainer on nuclear power by Hank Green which comes second in terms of views has structural significance as a policy-related video and got the highest number of comments on any individual video in the dataset (17,752), though its discussion strays far away from discussions around fuel prices and would certainly be captured as a separate cluster in the topic modeling exercise. Secondly, the Toyota Land Cruiser review video which occupies the first position on the table demonstrates the cost of wide queries.

5.4 Temporal Patterns and Event Detection

The most striking pattern in the entire dataset emerges when comment activity is plotted over time. Monthly comment volumes are summarised in Table 6 below for the period 2024 onwards .

Period	Comments	Notable Context	vs Baseline
2024 (full year, monthly avg)	32	Baseline period	1.0×
2025 (full year, monthly avg)	36	Baseline period	1.1×
Jan – Feb 2026	40	Pre-event baseline	1.2×
March 2026	3,680	Hike announcement	115×
April 2026	1,107	Post-event commentary	35×
May 2026 (partial)	3,230	Renewed coverage	101×

This difference could not be more striking. For the years 2024 and 2025, the monthly volume of comments on fuel-price content in our data set hovered around thirty to forty comments. The number skyrocketed to 3,680 in March 2026, a factor of almost one hundred and fifteen compared to base level, falling back to 1,107 in April but rising again to 3,230 in May 2026 (partial month of record up until 18 May), giving the same kind of double peak seen in the RUOK Day phenomenon, where a strong center event gives rise to a secondary tail of commentary and discussion. Given that in May the number of posts has already exceeded April's total mid-monthly, we can assume that this is still an ongoing event.

For event detection, a frequency ratio analysis would be more than enough to identify March and May 2026 clearly as event months; this is particularly relevant when analyzing subsequent topics and sentiment, since March marks a clear break in the conversation and the post-March period must be seen as a separate conversation.

Intra-week and intra-day patterns

At smaller intervals, comment activity displays two prominent cycles. In terms of day of the week, the highest number of comments (1,708) occurs on Fridays, followed closely by Mondays (1,610) and Sundays (1,557), whereas Tuesdays record the lowest number of comments (876)—about half that of Fridays. The Friday high may be due to weekend-anticipating news consumption behavior, but could also be explained by the Australian retail cycle of petrol sales, where fuel sales bottom out during the middle of the week and peak on Thursdays and Fridays, thus leading to discussions about prices on Fridays. In terms of hour of the day in UTC, the highest level (544) occurs at 21:00 UTC (which equals around 07:00 AEST—the next morning in Australia—when Aussies commute), with the second highest between 14:00 and 17:00 UTC (around morning time in New York and evening time in Australia).

5.5 Vocabulary Preview

Finally, in the context of qualitative checks performed during the data preprocessing phase, the frequency distribution of terms in the preprocessed comment corpus revealed that the dominant tokens (after excluding domain stopwords) fall into three categories: terms referring to issues of blame and accountability (government, politician, taxes), economic strain (money, pay, afford, working, family), and everyday experience (drive, car, week, day, work). Notably, the exclusion of fuel-related terms on the basis of domain stopwords proved to have been effective since none of the top tokens is related to fuel; nonetheless, the list serves as good confirmation of the quality of stopword choices made.

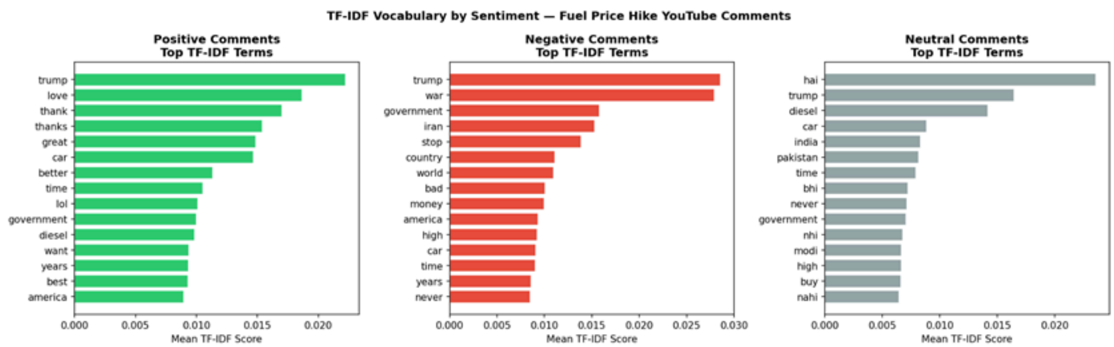

Figure 1. TF-IDF vocabulary usage by sentiments

It is important to note that the vocabulary review conducted in this case was informal; a more detailed TF-IDF and LDA analysis of the comment vocabulary will be presented further in this report. Nevertheless, preliminary insights gained through simple inspection suggest some expectations regarding the results of the LDA topic analysis, such as political and economic themes along with practical issues surrounding everyday costs of living. These topics correspond to what might be expected to arise within the context of a fuel price hike conversation.

5.6 Section Summary and Lead-In

Collectively, the early findings lead to the identification of four key facts that serve as the basis of the subsequent analysis. First, participation in discussions surrounding fuel prices follows a classic power law curve, with the majority of activity concentrated around a very small number of videos featuring particularly high levels of interaction. Second, the supply side of the discussion can be divided into three categories: Australian media, American media, and a significant contingent of Indian media organizations. The latter factor will have important consequences for drawing conclusions specific to Australia from the dataset. Third, there is a clear event-driven pattern of peak activity occurring during March and May 2026, with the latter still ongoing toward the end of the collection period. Finally, based on the language used, the most probable topics include themes of accountability, hardship, and experience.

The subsequent parts of the report are based upon these premises. Sentiment analysis will measure the emotions associated with the spike period relative to the more stable pre-period. Topic modeling will make the theme-based structure outlined above into something formalized. Network analysis will map out how the conversations actually occur through certain channels and communities of commenters. Each of the above approaches is essential but insufficient alone, as the worth of the entire report comes from combining all three.

6. Sentiment Analysis

Sentiment analysis was performed using the VADER (Valence Aware Dictionary and Sentiment Reasoner) framework. VADER was selected because it is specifically designed for analysing short-form social media content and is capable of handling informal language, punctuation emphasis, negation, and intensifiers commonly found in YouTube comments. Each comment received a compound sentiment score ranging from -1 to 1 and was subsequently classified as positive, neutral, or negative using standard VADER thresholds.

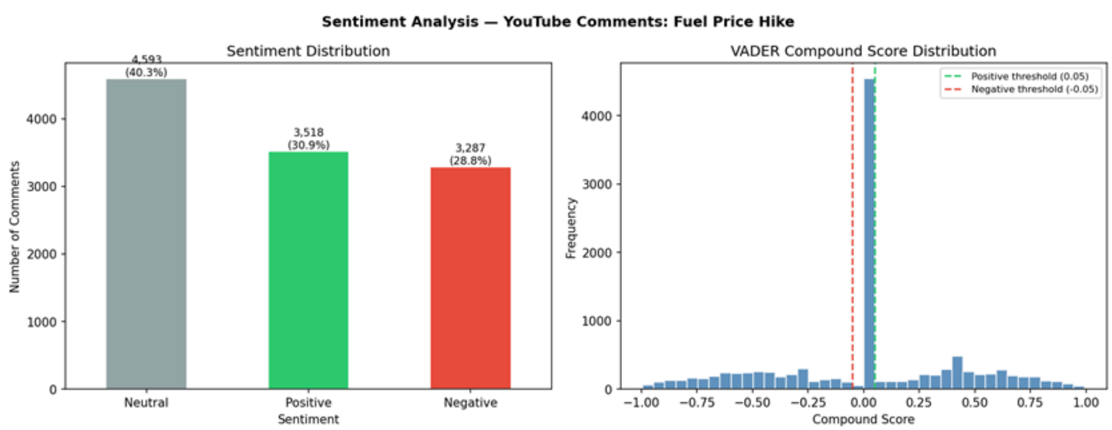

Figure 2. VADER score distribution with sentiments

The sentiment distribution demonstrates that neutral comments constituted the largest proportion of the dataset (40.3%), followed by positive comments (30.9%) and negative comments (28.8%). The predominance of neutral comments suggests that users were often engaging in discussion, information sharing, and opinion exchange rather than expressing strongly emotional reactions. This finding differs from crisis-driven social media discussions where negative sentiment frequently dominates.

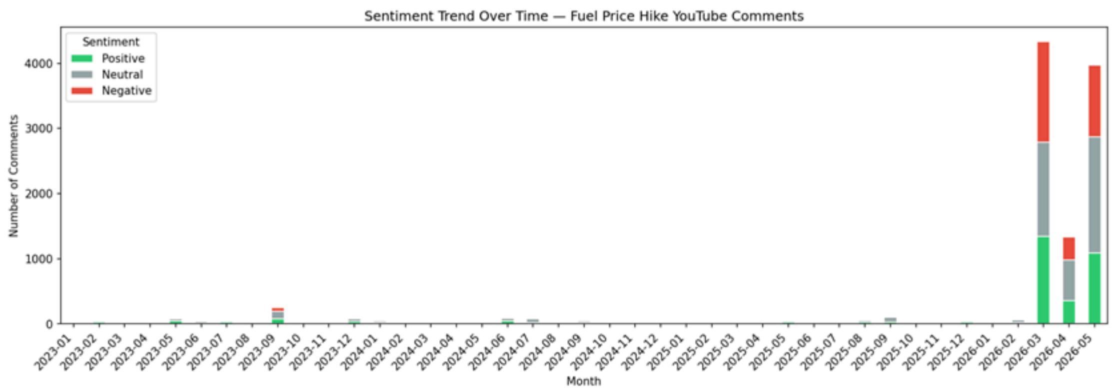

Figure 2. Youtube comments sentiment trend Over Time

The sentiment trend over time indicates fluctuations in public engagement, with periods of increased activity corresponding to heightened public interest in fuel price developments. Examination of sentiment-related vocabulary reveals that positive comments frequently contained words associated with approval and support, whereas negative comments were more likely to reference government actions, economic hardship, and geopolitical tensions.

This suggests that dissatisfaction was frequently directed towards broader political and economic circumstances rather than fuel prices alone.

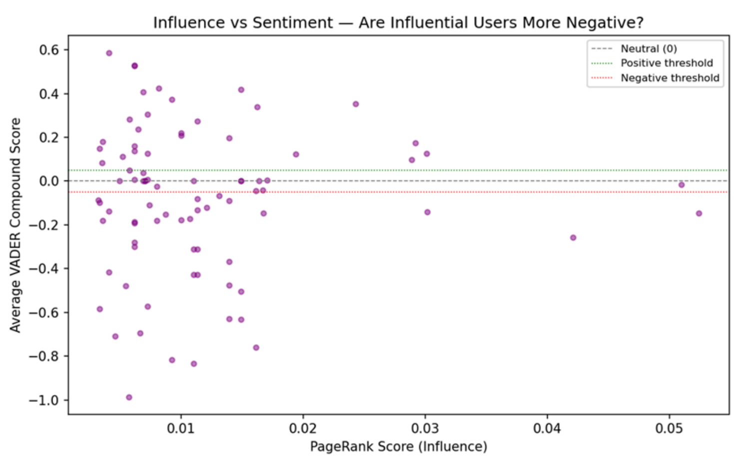

Figure 3. Influence vs Sentiment - Are they more negative?

The influence-versus-sentiment analysis further indicates that influential users participate across the entire sentiment spectrum. No strong relationship was observed between user influence and sentiment polarity, suggesting that network influence is driven primarily by participation and connectivity rather than a particular viewpoint.

7. Topic Modelling

Topic modelling was conducted using Latent Dirichlet Allocation (LDA) to uncover latent thematic structures within the YouTube comments. Prior to modelling, comments were tokenised, cleaned, and transformed into a document-term representation. Topic evaluation metrics indicated that a six-topic solution provided an effective balance between interpretability and model quality.

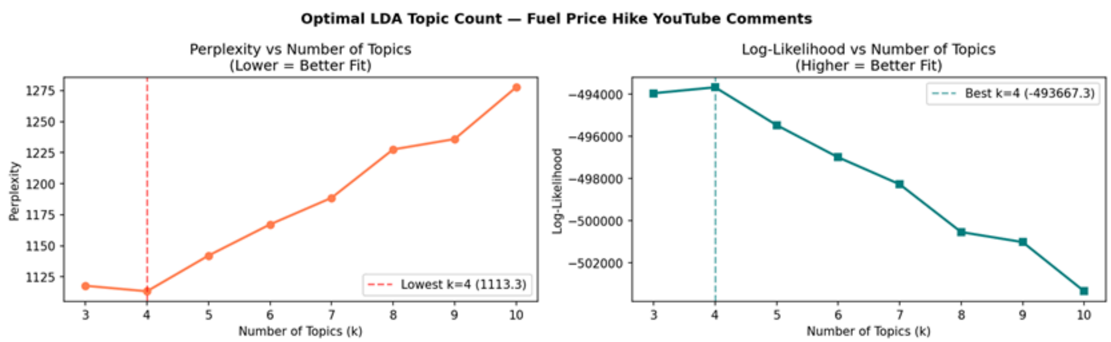

Figure 5. Optimal LDA topic counts

Analysis of the identified topics reveals that fuel-price discussions extend significantly beyond direct economic concerns. Several topics were dominated by terms such as “Trump”, “war”, and “Iran”, indicating that users frequently associated fuel price increases with geopolitical events and international conflicts. Other topics focused on government policy, transportation costs, fuel consumption, and national economic performance.

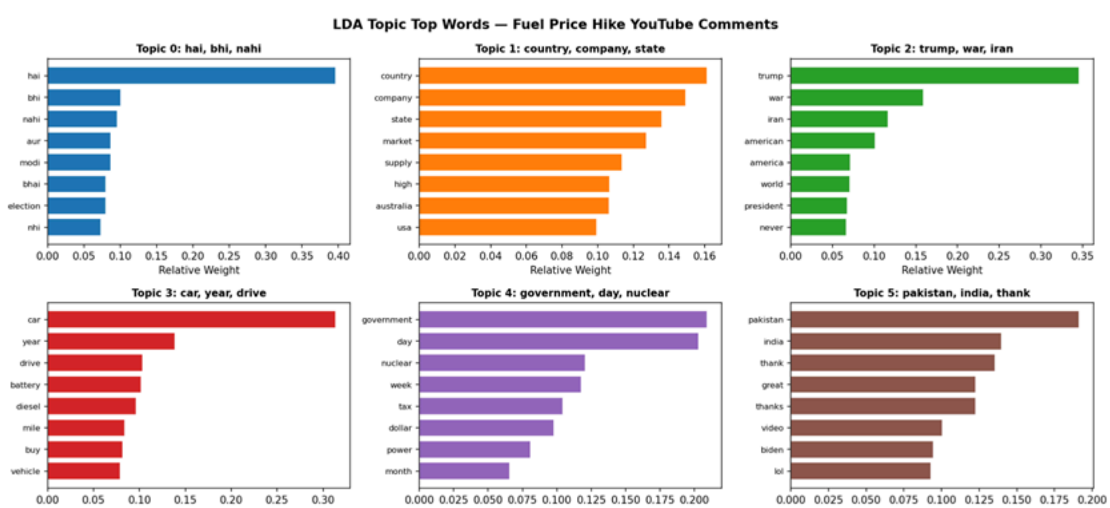

Figure 6. LDA topic top words usage from comments

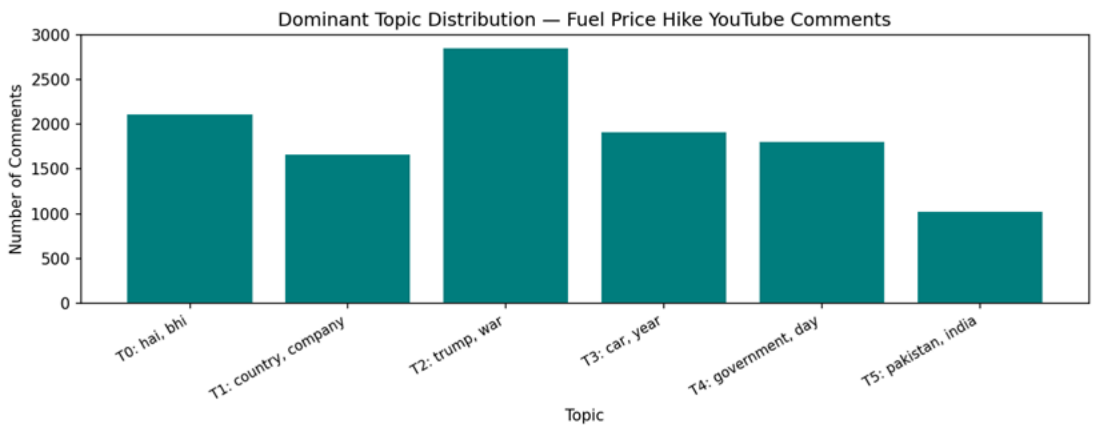

Figure 7. Dominant topic distribution on comments

The topic distribution analysis shows that geopolitical and political discussions formed a substantial proportion of the overall conversation. This suggests that users often interpret fuel price changes through broader political narratives rather than viewing them solely as market-driven phenomena. Such findings are consistent with prior social media studies demonstrating that public discourse frequently integrates economic events with political and ideological perspectives.

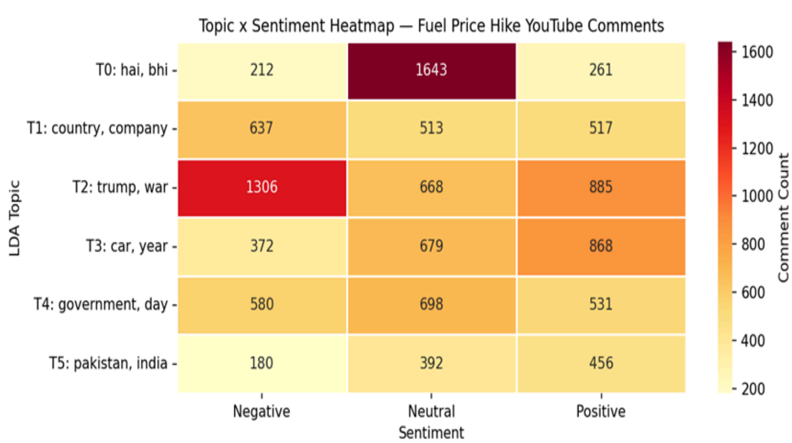

Figure 8. Sentiment heatmap on Negative, Neutral and positive comments

The topic-sentiment heatmap provides further insights into the emotional characteristics of each discussion theme. Political and geopolitical topics exhibited relatively higher levels of negative sentiment, whereas transportation-focused discussions contained a larger proportion of neutral and positive comments. These patterns indicate that sentiment is strongly influenced by the thematic context of the discussion.

8. Network Analysis

Network analysis was conducted to investigate patterns of interaction and influence among participants within the discussion ecosystem. A co-comment network was constructed in which nodes represented users and edges represented interaction relationships derived from commenting behaviour. Centrality measures including PageRank and betweenness centrality were calculated to identify influential participants and information brokers.

The results reveal a highly unequal network structure. A relatively small number of users occupy strategically important positions characterised by high PageRank and high betweenness centrality scores. These users serve as influential actors capable of amplifying information and shaping discussion visibility. Their positions enable them to connect otherwise separate communities, facilitating information flow throughout the network.

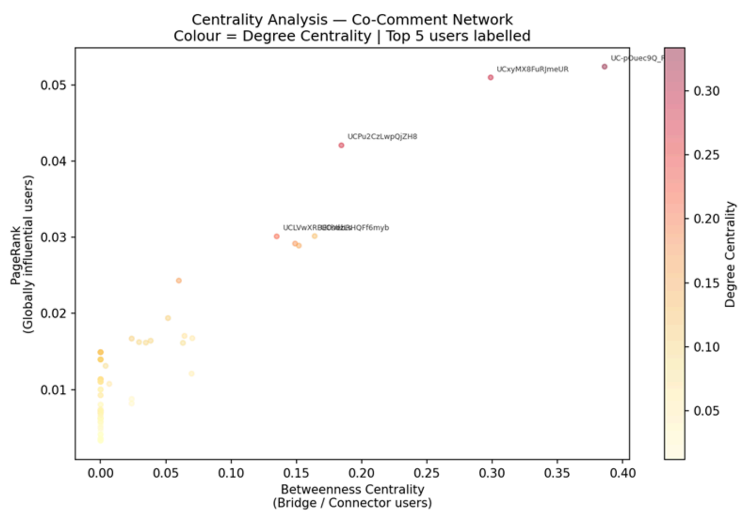

Figure 9. Centrality analysis on network with top 5 users

Betweenness centrality is particularly important because it identifies users acting as bridges between different clusters of participants. Information passing through the network is therefore disproportionately dependent on these users. As a result, they play a critical role in determining how narratives spread and evolve within the discussion environment.The centrality scatter plot highlights the existence of several highly influential users who simultaneously possess strong PageRank and betweenness centrality values. This combination indicates that these participants are not only well connected but also strategically positioned within the network structure. Such users may significantly influence the dissemination of opinions and information regarding fuel-price developments.

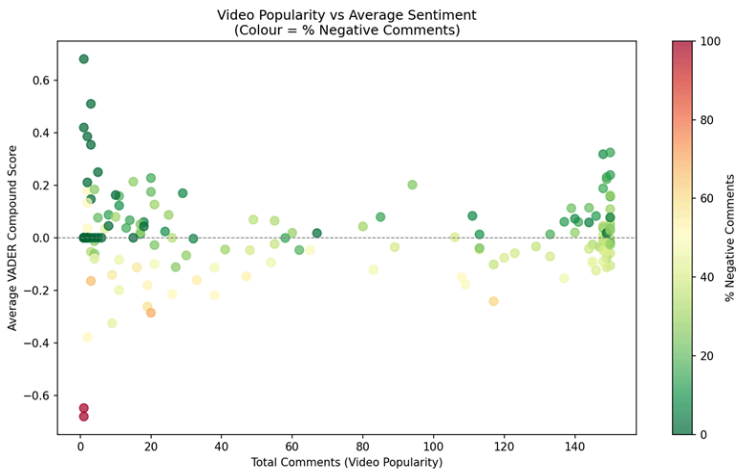

Figure 10. Video popularity vs Average Sentiment

The relationship between video popularity and sentiment was also examined. Although highly viewed videos generated larger volumes of engagement, no strong relationship was observed between popularity and average sentiment. Both positive and negative viewpoints appeared within highly popular videos, suggesting that audience engagement is driven primarily by topic relevance and controversy rather than sentiment polarity alone.

9. Discussion

The integration of sentiment analysis, topic modelling, and network analysis provides a comprehensive understanding of public discourse surrounding fuel-price increases. The results demonstrate that discussions are not dominated by a single viewpoint but instead encompass a diverse range of opinions, interpretations, and thematic concerns.

Sentiment analysis revealed a predominantly neutral discussion environment, indicating that users frequently engage in information sharing and debate. Topic modelling showed that fuel-price discussions are closely linked to broader issues including government policy, geopolitical conflict, economic management, and transportation costs. These findings suggest that social media users rarely interpret fuel-price increases as isolated economic events.

Network analysis further revealed that information dissemination is concentrated among a relatively small number of influential actors. These participants occupy strategically important positions within the communication network and play a significant role in shaping public discussion. Together, the findings indicate that fuel-price discourse on YouTube represents a complex socio-political conversation influenced by economic concerns, political narratives, and network dynamics.

10. Conclusion

This study employed social media analytics techniques to investigate public discussions relating to fuel-price increases on YouTube. By combining sentiment analysis, topic modelling, and network analysis, the study provided a multi-dimensional understanding of public opinion and communication dynamics.

The findings revealed that neutral comments represented the largest sentiment category, suggesting substantial information sharing and discussion activity. Topic modelling identified several dominant themes centred on geopolitical conflict, government policy, transportation issues, and economic concerns. Network analysis demonstrated that information dissemination is concentrated among a relatively small number of influential participants who act as information bridges across the network.Despite providing valuable insights, several limitations should be acknowledged. The study focused exclusively on YouTube comments, which may not fully represent broader public opinion.

In addition, the analysis was limited by API restrictions, language filtering processes, and the selected observation period. Future research could incorporate multiple social media platforms, longer collection periods, and advanced community detection techniques to further explore the evolution of public discourse surrounding fuel-price developments.

Overall, the results demonstrate the value of social media analytics for understanding how economic issues are discussed, interpreted, and disseminated within online communities.

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